

PKs, FKs, and Relationship Justifications

Table	Candidate Key(s)	Primary Key	Justification
TimePeriod	FL_DATE	FL_DATE	Each date uniquely determines year, month, day, day of week, and quarter.
Airlines	OP_CARRIER_AIRLINE_ID	OP_CARRIER_AIRLINE_ID	Airline ID uniquely identifies an airline.
AirlineNames	OP_CARRIER_AIRLINE_ID	OP_CARRIER_AIRLINE_ID	One airline has one official name.
Airports	AIRPORT_ID	AIRPORT_ID	Airport ID uniquely identifies an airport.
AirportNames	AIRPORT_ID	AIRPORT_ID	One airport has one official airport name.
StateNames	STATE_ABR	STATE_ABR	Each state abbreviation uniquely identifies a state.
Routes	(ORIGIN_AIRPORT_ID, DEST_AIRPORT_ID), ROUTE_ID	ROUTE_ID	A surrogate key was introduced to simplify references from the Flights table.
Flights	(OP_CARRIER_AIRLINE_ID, OP_CARRIER_FL_NUM, FL_DATE), FLIGHT_ID	FLIGHT_ID	A surrogate key uniquely identifies each flight occurrence and simplifies references.
FlightPerformance	FLIGHT_ID	FLIGHT_ID	Each flight has one performance record.
CancellationReason	CANCELLATION_CODE	CANCELLATION_CODE	Each cancellation code corresponds to one cancellation description.
DelayType	DELAY_ID	DELAY_ID	Each delay type has a unique identifier.
FlightDelay	(FLIGHT_ID, DELAY_ID)	(FLIGHT_ID, DELAY_ID)	One flight may have multiple delay types, but only one record per delay type.
FlightDiversion	(FLIGHT_ID, DIVERSION_NUMBER)	(FLIGHT_ID, DIVERSION_NUMBER)	One flight may have multiple diversion events identified by diversion sequence number.

Child Table	Foreign Key	Parent Table	Relationship Justification
Airports	STATE_ABR	StateNames	Each airport belongs to one state, while a state can contain many airports.
AirportNames	AIRPORT_ID	Airports	Each airport name belongs to exactly one airport.
AirlineNames	OP_CARRIER_AIRLINE_ID	Airlines	Each airline name belongs to exactly one airline.
Routes	ORIGIN_AIRPORT_ID	Airports	Every route must start at a valid airport.
Routes	DEST_AIRPORT_ID	Airports	Every route must end at a valid airport.
Flights	OP_CARRIER_AIRLINE_ID	Airlines	Each flight is operated by one airline, while an airline operates many flights.
Flights	ROUTE_ID	Routes	Each flight follows one route, while a route may be used by many flights.
Flights	FL_DATE	TimePeriod	Each flight occurs on one date, while many flights can occur on the same date.
FlightPerformance	FLIGHT_ID	Flights	Each performance record belongs to exactly one flight.
FlightPerformance	CANCELLATION_CODE	CancellationReason	A cancelled flight may reference a cancellation reason.
FlightDelay	FLIGHT_ID	Flights	A flight may have multiple delay records.
FlightDelay	DELAY_ID	DelayType	Each delay record must correspond to a valid delay category.
FlightDiversion	FLIGHT_ID	Flights	A flight may have multiple diversion events.
FlightDiversion	DIV_AIRPORT_ID	Airports	Each diversion airport must correspond to a valid airport.